# Ali Saeed Chaudhary

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## EDUCATION

**Iowa State University** Ames, Iowa
*B.S., Mechanical Engineering - Cyber-Physical Systems Minor.* ***GPA: 4.0.*** *May 2028*
**Korea University** Seoul, South Korea
*Study Abroad Program.* *June 2026*
Studied fundamentals of robotics and mechatronics, developed reinforcement learning algorithms using PPO, TRPO, and VLA Models for robotics applications, Gained coursework knowledge in nuclear power and nuclear energy

## EXPERIENCE

**Research Assistant** Aug 2026 – Present
*Iowa State University, Coordinated Systems Lab* *Ames, Iowa*

- Developing RNN-based Uncertainty-aware Calibrated Forecasting of Aircraft Trajectories at Scale in collaboration with RTX Corporation.
- Driving early-stage model development by researching flight trajectory datasets including NASA Sherlock data warehouse, building data-processing pipelines, and evaluating training approaches for air traffic prediction.
- Investigating techniques for modeling aircraft trajectory deviations and improving prediction reliability in dynamic airspace environments.

**Radiology IT Support Coordinator** Nov 2024 – Feb 2026
*Concord Medical Management, LLC* *Flower Mound, Texas (remote)*

- Served as a primary point of contact for radiologists and healthcare facilities, diagnosing and resolving time-sensitive technical issues affecting medical imaging workflows.
- Took ownership of troubleshooting and issue escalation, coordinating with facilities to restore system functionality.
- Streamlined communication and escalation workflows across departments, improving the efficiency and responsiveness of customer-facing IT support.

## ACTIVITIES & LEADERSHIP

**Cardinal Space Mining, NASA Lunabotics** Aug 2026 - Present
*Collections Sub-Team Member* *Ames, Iowa*

- Leading dust-cover design for the collections sub-system of a lunar regolith mining rover
- Improving robustness over the prior year's design while maintaining functionality and manufacturability

**Cyclone Rocketry, sponsored by Boeing** Oct 2024 - Dec 2025
*Mechanical Sub-Team Member* *Ames, Iowa*

- proposed and lead the designing of an airbrakes flight mechanics system
- modeled components in SolidWorks and ran FEA simulations in Ansys

## PROJECTS

**RL-Trained Robotic Hand** May – June 2026

- Trained a reinforcement learning policy in Isaac Sim to control a robotic hand using reference data from the HO-Cap (NeurIPS 2025) dataset
- Engineered a custom reward function prioritizing fingertip and palm contact, stable grasp formation, and reduced unnecessary motion.
- Configured policy observations using relative hand-object keypoints, joint states, and fingertip contact forces to guide learning.

**Substrate Rubbing Machine** May – August 2024

- Led design, construction, and testing of a custom substrate rubbing machine for the FMOD Research Lab, automating a previously manual process
- Designed the machine in Onshape, designed a custom PCB for the button panel, and 3D-printed control surfaces
- Implemented an automated rubbing routine with manual override controls; deployed to the lab after testing

**Private Home Server** June 2026 – Present

- Building a private media server for storing, hosting, and distributing services and files
- Configured remote access through an Oracle VPS using Pangolin and Traefik as a reverse-proxy, routing multiple services behind a single-entry point

## SKILLS

**CAD & Simulation:** SolidWorks, Onshape, Fusion360, Isaac Sim, Ansys FEA, CAM
**Programming & ML:** Python, RNNs, PPO, SAC, VLM / VLA Models, Robotics, ROS
**Systems & Tools:** Debian Linux, TailScale, Network Systems Management
**Manufacturing:** CNC Milling, Welding (TIG, MIG), Metal Working, 3D-Printing